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# Past, Present, and Future of Dexamethasone in Multiple Myeloma and AL Amyloidosis

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## ABSTRACT

For over half a century, dexamethasone has been a backbone of treatment regimens for plasma cell disorders such as multiple myeloma (MM) and light-chain (AL) amyloidosis. Dexamethasone doses of approximately 40 milligrams (mg) once weekly are often continued for months or years despite accumulating evidence that they can cause dose-dependent toxicities such as cataracts and infections. Acute dexamethasone-related toxicities such as insomnia or pedal edema, even if low-grade by clinical criteria, can significantly interfere with patient quality of life. In the past 5 years, several trials have demonstrated the efficacy and improved tolerability of corticosteroid-sparing regimens in MM. In this review, we discuss these and other studies to comprehensively assess the role of dexamethasone in the modern era. In newly diagnosed MM, robust data support planned dexamethasone discontinuation after one to two cycles in older and frailer patients. In the maintenance setting, the risk-benefit ratio of prolonged dexamethasone is unfavorable. While randomized trials have shown that once-weekly dexamethasone adds value within doublet regimens in relapsed/refractory MM, its contribution to triplet and quadruplet regimens is uncertain. Furthermore, data suggest that indefinite dexamethasone may actually limit the feasibility and efficacy of subsequent postprogression therapies. In AL amyloidosis, dexamethasone 40 mg once weekly for 6 months is excessive and may predispose patients to volume overload. In our review, we also discuss dexamethasone as a premedication for CD38-targeted monoclonal antibodies (where it is no longer required after one to two cycles) and for supportive care (where lower doses of 4-8 mg as needed can often suffice). Despite the historical inertia of corticosteroid-containing regimens in clinical trials and practice guidelines, corticosteroid-sparing regimens warrant prospective investigation across the gamut of plasma cell disorders.

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## INTRODUCTION

With one exception, the treatment of plasma cell disorders would be unrecognizable to an oncologist practicing as recently as two decades ago. Serial waves of novel therapeutic classes have extended progression-free survival (PFS) and overall survival (OS) across the spectrum of newly diagnosed multiple myeloma (NDMM), relapsed/refractory multiple myeloma (RRMM), and light-chain (AL) amyloidosis. Even the role of autologous stem-cell transplantation (ASCT), once incontrovertibly the standard of care for all eligible patients with NDMM or AL amyloidosis, is now being reappraised with the advent of quadruplet induction and measurable residual disease (MRD) assays.[1-3]

The exception to this constant state of evolution is dexamethasone, which has remained seemingly immortal in multiple myeloma (MM) and AL amyloidosis therapies for over 50 years. Dexamethasone and other corticosteroids undoubtedly have a role to play in the treatment of plasma cell disorders given their lymphotoxic properties and their roles in injection-/infusion-related reaction (IRR) prophylaxis and symptom management. However, corticosteroids are also associated with many short-term and long-term toxicities (Fig 1) that may affect patient quality of life (QOL) even if low-grade in nature.[4] Despite a mixed evidence base for dexamethasone from randomized controlled trials (RCTs) in the modern era (Table 1),[5-8,10-12] its unique historical inertia is visible across the spectrum of MM. In NDMM, dexamethasone is the only medication where a dosing schema previously shown to result in excess mortality—namely, 4-day pulses of 40 milligrams (mg) once daily—was subsequently reused in phase III trials a decade later.[6,13,14] In RRMM, dexamethasone is the only medication that is routinely recycled in new lines of therapy despite its clear failure in previous lines.

The concept of reducing dexamethasone dose intensities in trials has thus become appealing in recent years. However, there are several caveats to highlight with the burgeoning

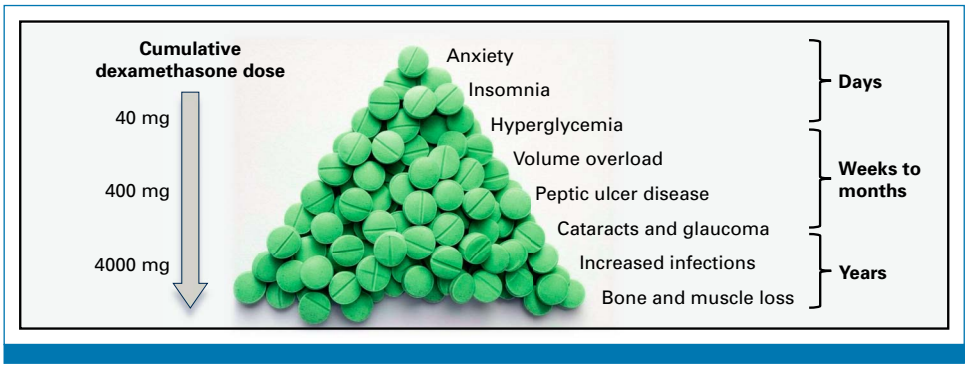


**FIG 1.** Selected toxicities of dexamethasone in MM and AL amyloidosis. Abbreviations: AL, light chain; mg, milligrams; MM, multiple myeloma.

"down with dex" movement.[15] Despite the success of corticosteroid-sparing schemas in ASCT-ineligible NDMM (Table 1),[5-7] these approaches have not been investigated prospectively in ASCT-eligible NDMM and have underperformed in RRMM.[10,11] As noted above, dexamethasone may also play key roles in IRR prevention, nausea prophylaxis, and bone pain management. When precisely to dose-reduce dexamethasone in MM and AL amyloidosis thus remains unclear, as does the margin of reduction required to optimize dexamethasone's benefit-risk ratio in the modern era. These unanswered questions—*who*, *when*, and *by how much*—are discussed in this review with an emphasis on areas of consensus versus controversy.

## DEXAMETHASONE: A PRIMER IN PLASMA CELL DISORDERS

A targeted history of corticosteroid-related observations and trials leading into the modern era is shown in Figure 2.[5,6,16-22] As early as 1950, authors of a case report concluded that "the

**TABLE 1.** Selected RCTs of Dexamethasone in Multiple Myeloma

| Trial | No. | Higher-Intensity Dexamethasone Arm | Corticosteroid-Sparing Arm | Efficacy of the Corticosteroid-Sparing Arm | Safety of the Corticosteroid-Sparing Arm |
|---|---|---|---|---|---|
| Newly diagnosed MM (transplant-ineligible) | | | | | |
| Facon 2006 (IFM 95-01)[5] | 488 | Dex pulses with or without other agents, eg, melphalan | Prednisone pulses plus melphalan | Superior PFS, comparable OS | Fewer toxicities |
| Rajkumar 2010 (E4A03)[6] | 445 | Lenalidomide plus dex pulses | Rd (once-weekly dex) | Superior PFS, superior OS | Fewer high-grade toxicities |
| Manier 2025 (IFM 2017-03)[7] | 295 | Rd continued until PD | Dara-len, with dex stopped after 2 months | Superior PFS, superior OS | Comparable toxicities |
| Postinduction MM therapy | | | | | |
| Larocca 2021[8] | 199 | Continuous Rd (with dex 20 mg once weekly) until PD | Len alone as maintenance | Comparable PFS, trend toward superior OS | Better tolerability, fewer discontinuations |
| Shustik 2007 (CTG MY.7)[9] | 292 | Monthly dex pulses as maintenance | Observation | Inferior PFS, comparable OS | Fewer infections and hyperglycemia |
| Relapsed/refractory MM | | | | | |
| Richardson 2014 (MM-002)[10] | 221 | Pom-dex | Pomalidomide monotherapy | Inferior ORR, inferior PFS, comparable OS | Comparable toxicities |
| Dimopoulos 2021[11] | 164 | Isatuximab plus dex | Isatuximab monotherapy | Inferior ORR, inferior PFS, comparable OS | Fewer psychiatric and GI toxicities |
| Kalff 2022 (ALLG MM-14)[12] | 78 | Pom-dex continued until PD | Pom-dex, with dex stopped after 4 months | Inferior PFS initially, but superior PFS/OS after 18 months[a] | Fewer infections and less neuropathy |

NOTE. Details of each trial are heavily simplified for brevity. Where mentioned, dex pulses refer to 40 mg once daily for 4 consecutive days. Abbreviations: Dara, daratumumab; dex, dexamethasone; len, lenalidomide; MM, multiple myeloma; ORR, overall response rate; OS, overall survival; PD, progressive disease; PFS, progression-free survival; Pom-dex, pomalidomide/dexamethasone; RCT, randomized controlled trial; Rd, lenalidomide/dexamethasone.

[a]Superior landmarked PFS and OS in preplanned analyses with adjustment for multiplicity of comparisons. See the text for details.

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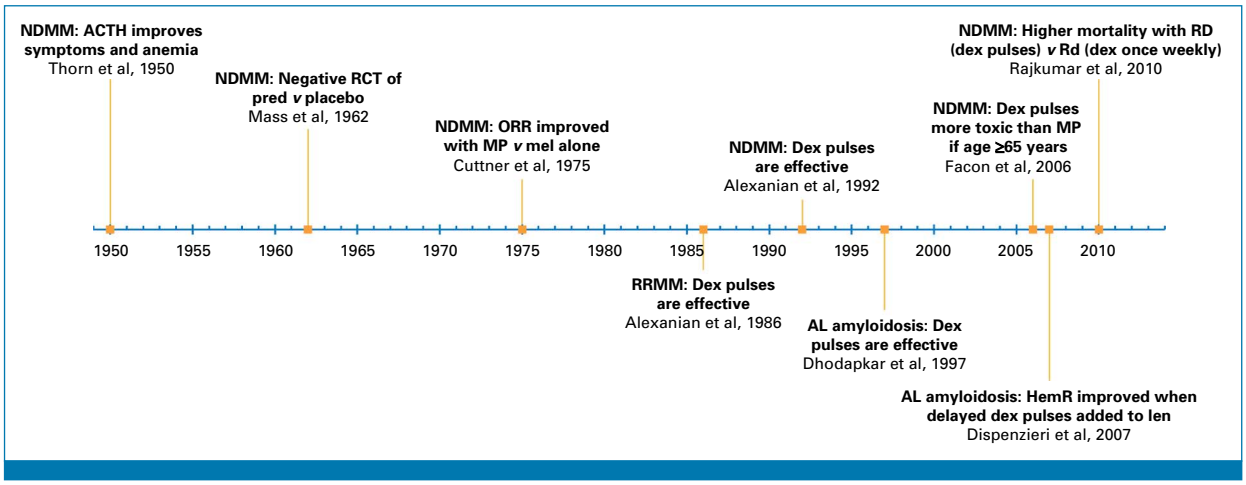


**FIG 2.** History of corticosteroids in MM and AL amyloidosis. This figure illustrates representative studies from the early 1950s leading up to the ECOG E4A03 trial, which was published in 2010.[6] The details of each study are heavily simplified for brevity; refer to the original publications for details.[5,6,16-22] ACTH, adrenocorticotropic hormone; AL, light-chain; Dex, dexamethasone; HemR, hematologic responses in AL amyloidosis, len, lenalidomide; mel, melphalan; MM, multiple myeloma; MP, melphalan/prednisone; NDMM, newly diagnosed multiple myeloma; ORR, overall response rate; RCT, randomized controlled trial; RD, lenalidomide with dexamethasone pulses; Rd, lenalidomide with once-weekly dexamethasone; RRMM, relapsed/refractory multiple myeloma.

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reaction of [myeloma] cells to the adrenal steroids is apparently strikingly similar to that of lymphocytes; and the marked fall in abnormal serum globulin follows the pattern of the presumed source—that is, the myeloma cell."[16] While prednisone was the predominant choice of therapy at first, 4-day dexamethasone pulses gained traction by the late 1990s for both MM and AL amyloidosis.[19-21,23-25] As newer therapies emerged, 4-day dexamethasone pulses served as the control arms of phase III trials establishing the superiority of drugs like thalidomide, lenalidomide, and bortezomib.[26-29] However, the multiarm phase III IFM 95-01 trial of patients with NDMM found that fatal adverse events (AEs) from infections and GI bleeding were much more common in arms containing dexamethasone pulses.[5] Soon afterward, the phase III ECOG E4A03 trial demonstrated increased mortality with 4-day pulses of dexamethasone 40 mg once daily plus lenalidomide as opposed to modern Rd containing once-weekly dexamethasone 40 mg.[6]

In light of the ECOG E4A03 trial's results, most subsequent trials have shifted away from dexamethasone pulses. This is evidenced in the Data Supplement (Table S1, online only), which tabulates all phase III trials in plasma cell disorders published within the past 5 years. Even within this narrow timeframe, however, there remains significant heterogeneity in how dexamethasone is prescribed. Twelve of 39 modern-era trials have used cycle 1 dexamethasone schedules that exceeded (on average) 40 mg once weekly, including two trials with 4-day pulses. Conversely, there is no discernable pattern to which trial protocols incorporate versus omit provisions for initial dose reductions based on age ≥75 years and/or low BMI (Data Supplement, Fig S1).

## The Science of Dexamethasone and Its Toxicities

The various mechanisms of action of corticosteroids in plasma cell disorders, both directly and in synergy with other MM medications, have been described previously.[30-35] Dexamethasone's anti-inflammatory properties come from its analogous effects on normal lymphocytes and explain its use for IRR prophylaxis. While a full discussion of dexamethasone with chimeric antigen receptor T-cell (CAR-T) therapy is outside the scope of this review, dexamethasone is often used judiciously to manage cytokine release syndrome (CRS) and related toxicities given its ability to inhibit T lymphocytes without compromising CAR-T efficacy.[36,37]

Conversely, dexamethasone is a potent synthetic corticosteroid. As such, its cumulative toxicities (Fig 1) are similar to what one might expect in Cushing's syndrome or diabetes.[19,20,31,38-40] Unfortunately, specific research into the AE profile of dexamethasone in plasma cell disorders is limited. While several RCTs of once-weekly dexamethasone versus no dexamethasone have been conducted (Table 1),[7,8,10-12] most publications have reported only grade 3+ AEs (which thankfully are uncommon with once-weekly dexamethasone). AE data from two RCTs that did report any-grade AEs are summarized in Figure 3.[11,12] Even without formal hypothesis testing, it is clear to the reader that dexamethasone 20-40 mg once weekly is not entirely benign: for example, 25% any-grade insomnia with isatuximab plus dexamethasone versus 2% with isatuximab alone.[11] Other RCTs have elucidated dexamethasone-related toxicities more indirectly, most notably the negative GEM-CLARIDEX trial of Rd with or without

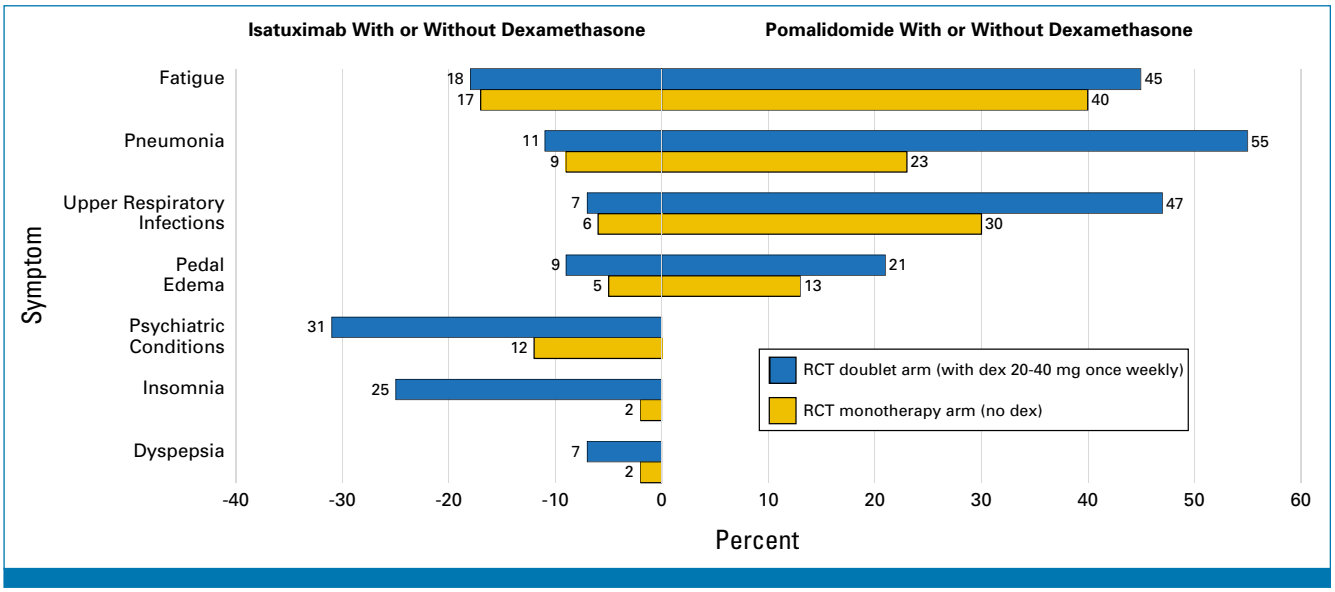


**FIG 3.** Selected all-grade AEs of dexamethasone from randomized trials. The trial labeled "Isatuximab With or Without Dexamethasone" was a phase II RCT of isatuximab with or without added dexamethasone (40 mg once weekly if age <75 years, 20 mg once weekly if age ≥75 years) in patients with RRMM and ≥3 previous LOT.[11] Any-grade AEs for the isatuximab + dexamethasone doublet arm (n = 55) and isatuximab monotherapy arm (n = 109) were assessed using CTCAE v4.03. The trial labeled "Pomalidomide With or Without dexamethasone" was a phase II RCT of pomalidomide with or without discontinuation of dexamethasone 40 mg once weekly after four cycles in patients with RRMM and ≥2 previous LOT.[12] Any-grade AEs for the pomalidomide + dexamethasone doublet arm (n = 38) and pomalidomide monotherapy arm (n = 40) were assessed using CTCAE v4.0. All percentages are derived from the original publications; no formal hypothesis testing was performed. AEs visualized here were chosen from the list of common corticosteroid-related toxicities shown in Figure 1 if AE data were available for one or both trials. AE, adverse event; CTCAE, Common Terminology Criteria for Adverse Events; dex, dexamethasone; mg, milligrams; LOT, line of therapy; RCT, randomized controlled trial; RRMM, relapsed/refractory multiple myeloma.

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clarithromycin in NDMM. The paradoxical increase in infectious deaths in the antibiotic-containing arm was directly attributed to clarithromycin-mediated CYP3A4 inhibition, leading to increased dexamethasone exposure above the nominal dose of 20 mg once weekly received by most patients.[41]

These findings are buttressed by real-world evidence (RWE) from a variety of MM settings, which have implicated dexamethasone in the development of several other toxicities: for example, insomnia, cataracts, adrenal insufficiency, and infections.[40,42-46] The negative impacts of chronic corticosteroids on bone health, glycemic control, and peptic ulcers have not been studied specifically in MM to our knowledge. However, several systematic reviews—both in healthy adults and in patients with autoimmune conditions—have identified such associations even with low corticosteroid dose intensities.[47-51] Finally, for understanding the patient experience with regard to dexamethasone-related AEs in MM, the Steroid Symptom Questionnaire-Multiple Myeloma (SSQ-MM) PRO inventory has recently been developed specifically for this purpose. In the validation study of this instrument, higher SSQ-MM scores were strongly associated with lower QOL and worsened psychological well-being in a diverse cohort of 140 patients.[4]

## The Logistics of Dexamethasone Dosing

Dexamethasone is generally dosed orally (PO), even for patients who are concurrently receiving parenteral treatment for MM or AL amyloidosis. Dexamethasone has excellent bioavailability, and studies from other malignancies suggest that PO dexamethasone outperforms intravenous (IV) dexamethasone as a premedication to prevent IRRs (likely because of its slower absorption and availability).[52,53] Indeed, the half-life of PO dexamethasone is 3-4 hours, but its biological effects can last for up to 2 days.[54] When dexamethasone is required as a premedication for patients who are deemed reliable, self-administered dexamethasone earlier in the day at home can lower dexamethasone-related insomnia and the overall time toxicity of treatment.[38,55] By contrast, IV dexamethasone is generally reserved for emergencies such as acute cord compression or high-grade CRS.

There is no consensus on whether strict once-weekly dosing versus split dosing—for example, 20 mg daily for 2 consecutive days rather than 40 mg at once—is a superior approach in plasma cell disorders. Certainly, split dosing can be considered on a case-by-case basis for patients at higher risk of AEs such as hyperglycemia. Similarly, patients who

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develop let-down symptoms (eg, fatigue and dizziness) with strict once-weekly administration can be transitioned to split dosing.[38] Interestingly, one *post hoc* analysis of trial-enrolled patients receiving twice-per-week bortezomib suggested that split-dosing dexamethasone alongside each bortezomib dose (eg, days 1, 2, 4, 5, 8, 9, 11, 12) was associated with less neuropathy than strict once-weekly dosing: any-grade incidence 46% versus 64% and grade 3+ neuropathy incidence 5% versus 11%.[56] However, this complex dosing schedule may not be applicable to once-weekly bortezomib schedules, which have a lower neuropathy risk and are used far more frequently in clinical practice.[57,58]

## DEXAMETHASONE IN INDUCTION FOR NDMM

### ASCT-Ineligible Patients

As shown in the Data Supplement (Table S1), all modern-era phase III trials of ASCT-ineligible NDMM have used average total dexamethasone doses below 60 mg weekly; encouragingly, half of these trials have also incorporated individualized dosing strategies (eg, 20 mg once weekly if age ≥75 years). It has become increasingly clear that prolonged full-intensity dexamethasone is unnecessary for these patients. For example, a pooled analysis of data from two trials analyzed over 500 patients who received dexamethasone totals of 40-60 mg weekly with Rd, bortezomib plus Rd (VRd), or elotuzumab plus VRd. Only 31% of patients were able to maintain their assigned dexamethasone doses for the entirety of the 24-week induction period; furthermore, there were no significant PFS or OS differences between patients who did versus did not maintain full-dose dexamethasone.[59]

More evidence for a corticosteroid-sparing approach comes from the recently published phase III IFM 2017-03 trial. This trial compared a modified version of daratumumab plus Rd (Dara-Rd) against Rd, with the Dara-Rd arm receiving only 8 weeks of dexamethasone 20 mg once weekly. By contrast, the Dara-Rd arm of the earlier phase III MAIA trial had received dexamethasone 20-40 mg once weekly until progression.[60] Any cross-trial comparisons should be approached with caution, given that IFM 2017-03 participants were much older and frailer than their MAIA counterparts (as detailed in the Data Supplement, Table S2). However, the relatively comparable PFS distributions with the two Dara-Rd approaches are striking—particularly given that cumulative dexamethasone exposure in IFM 2017-03 was <5% the corresponding figure for MAIA.[7,60] IFM 2017-03 showed a median PFS over twice as long with corticosteroid-sparing Dara-Rd versus traditional Rd: median PFS 53.4 months versus 22.5 months, hazard ratio (HR) 0.51, 95% CI, 0.37 to 0.70. Although MAIA results were reported before IFM 2017-03 results, the long-term risks of corticosteroids make the IFM 2017-03 dosing schema an attractive and evidence-based standard of care for ASCT-ineligible patients.

This specific approach is further buttressed by the recently published phase II REST trial, a single-arm trial of isatuximab plus VRd (Isa-VRd) quadruplet induction in ASCT-ineligible patients.[61] While several trials of Isa-VRd have been conducted in ASCT-ineligible patients (as detailed in the Data Supplement, Table S3),[61-63] the REST trial stands out in terms of its patient population: 31% of enrolled patients were 80 years and older, and 45% were considered frail. As with IFM 2017-03, dexamethasone 20 mg once weekly was stopped after 8 weeks. QOL was preserved longitudinally with this approach, and relative dose intensities for isatuximab and bortezomib both exceeded 95%.[61] While difficult to compare results directly against IMROZ and BENEFIT given a much older and frailer patient population, the 100% overall response rate (ORR) and 41% MRD negativity rate seen in the REST trial are quite impressive. This is doubly true given that cumulative dexamethasone exposure in the REST trial was <5% that of IMROZ and <25% that of BENEFIT. The BENEFIT trial has already demonstrated that Isa-VRd with once-weekly bortezomib can outperform a MAIA-style triplet.[62] The REST trial provides a blueprint (namely, with dexamethasone 20 mg once weekly for 8 weeks) to expand quadruplets to a much larger population.

### ASCT-Eligible Patients

How well this paradigm applies for ASCT-eligible patients is uncertain. As noted in the Data Supplement (Table S1), all modern phase III trials of ASCT-eligible NDMM have used total dexamethasone doses of ≥40 mg weekly. The general rationale for maintaining dexamethasone in this range is that younger patients are simultaneously more likely to both (1) benefit from its anti-MM effects and (2) have fewer QOL impairments from its AEs. However, it is unclear how true either of these statements are. For the first statement, the corticosteroid-sparing regimens described above demonstrate that prolonged dexamethasone is largely unnecessary in NDMM in general. If anything, the 41% MRD negativity rate from the REST trial would be the floor for how well an analogous quadruplet might function in younger and more fit patients. Furthermore, for many ASCT-eligible patients who remain MRD-positive after quadruplet induction, high-dose melphalan can deepen responses thereafter.[64,65]

For the second statement, there is no biological basis for dexamethasone's acute toxicities (eg, irritability and insomnia) being any less burdensome in younger versus older patients. One study using wearable devices to track sleep patterns during MM induction found that dexamethasone-related insomnia was actually more pronounced in younger patients (65 years and younger) than in older patients: 81.4 minutes/day versus 37.4 minutes/day on average, respectively.[42] For dose-dependent toxicities such as cataracts,[40] younger patients may be at a disproportionately increased lifetime risk compared with age-matched controls given their higher and more prolonged dexamethasone dose intensities. However, more prospective data are needed before such statements can be established with certainty.

## DEXAMETHASONE IN POSTINDUCTION THERAPY

Postinduction therapy, which includes both post-transplantation maintenance and eventual treatment de-escalation in ASCT-ineligible patients, can last for many years. In this setting, the cumulative toxicities of dexamethasone (Fig 1) almost certainly outweigh their benefits. The best example of this principle comes from a phase III trial of Rd in NDMM where dexamethasone 20 mg once weekly was either continued versus stopped after 9 months.[8] The corticosteroid-sparing arm met the study's primary end point for superior event-free survival, which included high-grade AEs or AEs requiring lenalidomide discontinuation. While PFS was comparable between the two approaches, there was an unmistakable trend toward improved OS with dexamethasone discontinuation: 3-year OS 74% versus 63%: HR 0.62, 95% CI, 0.37 to 1.03 ($P$ = .06). Despite a protocol-defined 60% lenalidomide dose reduction after 9 months at the same time as dexamethasone discontinuation, patients in the corticosteroid-sparing arm also ultimately received a higher cumulative lenalidomide dose as well. Evidently, discontinuing dexamethasone may help mitigate fatigue, infections, and other AEs that might otherwise have prompted lenalidomide dose reductions or interruption.

Although older RCTs (both published before 2010) demonstrated improved PFS with corticosteroid maintenance when used as monotherapy,[9,66] a more modern phase III trial showed no PFS benefit with the addition of prednisone to lenalidomide maintenance.[67] Conversely, in one older RCT, even only 4 days of dexamethasone per month (albeit dosed as a pulse) was associated with additional hyperglycemia and infections versus observation.[9] Based on the totality of evidence, dexamethasone should not be routinely used as a component of maintenance. Even for patients with ultrahigh-risk disease or plasma cell leukemia—that is, scenarios where patients may deem a higher AE burden to be acceptable—the phase II MUKnine/OPTIMUM trial demonstrated that dexamethasone discontinuation after 6 months of daratumumab plus VRd (Dara-VRd) maintenance was effective as part of a longitudinal intensive treatment strategy.[68]

## DEXAMETHASONE IN RRMM

Unlike in other settings, RCTs in RRMM have demonstrated ORR and PFS benefits with the addition of once-weekly dexamethasone to isatuximab monotherapy or to pomalidomide monotherapy (Table 1).[10-12] Given that pomalidomide-dexamethasone and isatuximab-dexamethasone doublets are no longer used in practice, the relevance of these findings to modern practice is unclear. Many of these PFS benefits appear to have been driven largely by improved ORR rather than by longer durations of response, suggesting that dexamethasone could potentially be discontinued once a response is achieved.[12] Indeed, the phase II ALLG MM-14 RCT investigated pomalidomide-dexamethasone therapy with or without the discontinuation of dexamethasone 40 mg once weekly after 4 months.[12] This trial demonstrated improved PFS with the continued-dexamethasone arm: median PFS 5.7 months versus 2.6 months (HR, 0.63 [95% CI, 0.40 to 1.00]). However, postprogression PFS favored the corticosteroid-sparing arm: median PFS 12.7 months versus 4.6 months (HR/CI not reported, $P$ = .03). Although the overall OS benefit for a corticosteroid-sparing approach did not reach statistical significance (median 25.7 *v* 17.4 months, HR, 1.36 [95% CI, 0.70 to 2.64]), several multiplicity-adjusted landmarked analyses after the first year of treatment did. In other words, among patients who remained alive for at least 1 year, OS analyses significantly favored patients whose dexamethasone had already been discontinued.[12]

Despite its lack of power to detect OS differences, the visible crossing of OS curves seen in the ALLG MM-14 trial suggests that time-limited dexamethasone following initial response warrants closer investigation. Despite the existence of several pathways known to mediate corticosteroid resistance, dexamethasone is the only drug in MM that is routinely recycled in subsequent lines of therapy.[30,31] Planned dexamethasone discontinuation after achievement of a sufficiently deep response could potentially prevent such corticosteroid-refractory clones from forming—which, in turn, could allow for the judicious reintroduction of dexamethasone at relapse to recapture responses. Conversely, accumulating toxicities (most notably infections and impaired bone/muscle health) can potentially limit patient candidacy for subsequent treatments at relapse if dexamethasone is continued indefinitely. However, these theories remain speculative at current time in the absence of dedicated analyses of drug sensitivity and patient fitness.

## DEXAMETHASONE IN AL AMYLOIDOSIS

Because there is considerable overlap between the symptoms of AL amyloidosis and the toxicities of dexamethasone, the use of corticosteroids in this setting requires additional caution. Underlying cardiac or renal involvement can predispose patients to develop fluid overload even before treatment initiation, which explains the presence of pedal edema as a key symptom driving PROs in AL amyloidosis.[69] Unfortunately, pedal edema is also a well-known toxicity of corticosteroids.[38] As detailed in the Data Supplement (Table S4), recent phase III trials of dexamethasone-containing regimens have demonstrated that any-grade edema occurs in over a third of patients with AL amyloidosis; grade 3 edema, which affects self-care and other activities of daily living, occurs in 3%-6% of patients.[70-72] For context, trials of analogous regimens for MM have not even reported edema as an AE at all.[73,74]

However, dexamethasone continues to be a backbone of treatment regimens for AL amyloidosis. This is in part based

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on extrapolation from the MM literature and in part based on older RWE, showing higher ORR with higher cumulative dexamethasone doses.[75,76] Given the rarity of AL amyloidosis, an RCT of corticosteroid-sparing strategies is unlikely to ever be performed in this setting. That being said, it is very likely that dexamethasone 20-40 mg once weekly for 6 months (as implemented in the ANDROMEDA trial)[70] is excessive in the modern era. Most cases of AL amyloidosis are driven by premalignant clones that are inherently more susceptible to shorter courses of clone-directed therapies. For example, daratumumab monotherapy is well tolerated and can lead to rapid hematologic responses even in advanced cardiac amyloidosis.[77] Even in the setting of relapsed/refractory AL amyloidosis, the phase III TOURMALINE-AL1 trial demonstrated that dexamethasone 20 mg once weekly (alongside ixazomib) can meaningfully delay organ deterioration.[71] We posit that dexamethasone 10-20 mg once weekly for one cycle with discontinuation thereafter would suffice in the modern era to induce hematologic responses and minimize toxicities; however, data to validate this approach are currently lacking.

## DEXAMETHASONE FOR SUPPORTIVE CARE

Corticosteroids are also widely used for IRR prophylaxis in both MM and AL amyloidosis. Given the historical inertia of dexamethasone doses ≥40 mg once weekly as an antineoplastic drug, research into the minimal effective dose and duration of dexamethasone necessary for other purposes has been neglected. That being said, montelukast works remarkably well to prevent IRRs with the first dose of CD38-targeted monoclonal antibodies such as daratumumab or isatuximab.[78-83] As evidenced by the IFM 2017-03 and REST trials, dexamethasone can be stopped after 8 weeks without recurrent IRRs; in fact, RWE suggests that dexamethasone can be stopped even earlier as a premedication.[7,61,84,85] With carfilzomib, IRRs can manifest with delayed flu-like symptoms or nausea/vomiting. However, trials of carfilzomib and carfilzomib-lenalidomide maintenance have demonstrated that lower doses (ie, dexamethasone 4-8 mg with each carfilzomib dose) can be used with success for IRR prophylaxis.[86,87]

Dexamethasone is also used for other purposes in MM: bone pain, prevention of pain after radiation therapy (RT), management of nausea and vomiting, and more. For focal RT to bone-based lesions, phase III data support the use of brief dexamethasone regimens (ie, a 5-day course of 4-8 mg once daily) to prevent pain flare thereafter.[88] With selinexor-containing regimens, expert recommendations suggest the continuous use of dexamethasone as an antiemetic regardless of response achieved.[89] However, in studies of selinexor for non-Hodgkin's lymphoma, dexamethasone 4-8 mg with each selinexor dose has sufficed for this purpose.[90,91] In general, the minimum effective dose rather than the maximum tolerated dose should be used with corticosteroids for supportive care in MM and AL amyloidosis.

## DISCUSSION

Few would disagree with the assertion that dexamethasone doses to treat plasma cell disorders are generally too high in the modern era. Corticosteroid-sparing triplets and quadruplets have demonstrated impressive results as induction regimens,[7,61] and patients who maintain full-dose dexamethasone during induction do not derive any downstream PFS or OS benefits as a result.[59] In RRMM, RCTs of indefinite once-weekly dexamethasone have demonstrated increased AEs (Fig 3) and questionable long-term outcomes even if PFS is initially improved.[10-12] In AL amyloidosis, volume overload exacerbated by once-weekly dexamethasone is a substantial driver of patient QOL.[69] If dexamethasone were a novel drug being investigated today, there is no doubt that its maximum tolerated dose would fall well below 40 mg once weekly.

A more challenging question is how specifically to operationalize this "down with dex" mantra.[15] Dexamethasone is an important component of treatment for plasma cell disorders, both directly given its lymphotoxic properties and also indirectly given its roles in symptom control and IRR prophylaxis. For patients with MM-related emergencies, dexamethasone pulses are absolutely appropriate as a rapidly acting and rapidly available therapy. Conversely, for patients on long-term maintenance therapy, the benefit-risk ratio of dexamethasone is almost certainly unfavorable. For patients in between these two extremes, corticosteroid dosing should be individualized. In Table 2, we summarize some of our suggested best practices for the use of dexamethasone in plasma cell disorders (also depicted visually in Fig 4 for MM). While only a minority of these suggestions come with a high level of evidence-based certainty, we hope that their dissemination can reduce some of the disparities in care that can stem from a lack of familiarity with dexamethasone's many nuances in this setting.[92-95]

Beyond the general lack of high-quality evidence, our review has other limitations as well. The myriad uses of dexamethasone for AE management during bispecific antibodies or CAR-T therapies, for example, CRS management or the attenuation of peri–CAR-T lymphocytosis, are outside the scope of this review.[96-99] Given our focus on CD38-targeted monoclonal antibodies with regard to IRR prophylaxis, we did not discuss the complexities of dexamethasone premedication before other parenteral MM medications such as elotuzumab.[100] Finally, we intentionally did not discuss nondexamethasone corticosteroids such as prednisone or methylprednisolone in MM and AL amyloidosis. These agents are reasonable both for MM control and for IRR prophylaxis, as summarized in the Data Supplement (Table S5).

Regardless, there is no doubt that corticosteroid dose intensities have shifted downward from an era where 22% of patients with RRMM treated with dexamethasone developed

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**TABLE 2.** Suggested Best Practices for Dexamethasone in Plasma Cell Disorders

| Recommendation | Certainty | Abbreviated Rationale |
|---|---|---|
| Dexamethasone pulses should generally be reserved only for emergencies, eg, cord compression or acute cast nephropathy | High | The E4A03 trial showed a significant reduction in treatment-related mortality with once-weekly dexamethasone versus dexamethasone pulses (40 mg once daily for 4 consecutive days) in NDMM[6] |
| Dexamethasone should generally be dosed orally, with total doses of 20-40 mg weekly (split dosing can be considered if needed) | Moderate | PO dexamethasone has excellent bioavailability and fewer toxicities. In some cases, split dosing (eg, 20 mg on 2 consecutive days instead of 40 mg) can reduce toxicities such as hyperglycemia[38] |
| In ASCT-ineligible NDMM regimens containing a CD38-targeted mAb, dexamethasone should be discontinued after two cycles | High | The IFM 2017-03 and REST trials have demonstrated the efficacy of this approach (dexamethasone 20 mg once weekly for 8 weeks, then discontinued) with triplets and quadruplets, respectively[7,61] |
| In ASCT-eligible NDMM, dexamethasone dose reduction or discontinuation should be considered if any-grade AEs develop | Low | Dexamethasone-related AEs can significantly impair QOL, even if low grade.[4] Certain AEs may be more burdensome or problematic for younger patients, eg, insomnia and cataracts.[40,42] |
| For maintenance, dexamethasone should be omitted altogether or (if deemed necessary for high-risk disease) discontinued after ≤6 cycles | High | Adding corticosteroids to lenalidomide maintenance does not improve PFS and may increase high-grade AEs.[8,67] Even in ultra-high-risk disease, dexamethasone has successfully been stopped after six cycles[68] |
| In RRMM, dexamethasone dose reduction or discontinuation should be considered after ≥VGPR or response plateau | Low | PFS benefits have largely been driven by ORR improvements.[10-12] Cumulative AEs from indefinite dexamethasone (eg, impaired muscle health) can limit candidacy for treatments at relapse |
| In AL amyloidosis, dexamethasone should be started at 10-20 mg once weekly regardless of age and discontinued after the first cycle | Low | Dexamethasone can substantially worsen fluid retention and edema. Hematologic responses are brisk with daratumumab,[70,77] and dexamethasone may be dispensable with these regimens |
| Further premedication with dexamethasone is not required with CD38-targeted mAbs if no IRRs occur after 1-2 cycles | Moderate | With daratumumab, eliminating all premedications is safe after the first cycle.[84,85] In both IFM 2017-03 and REST, dexamethasone was safely stopped after two cycles without a rise in IRR incidence[7,61] |

NOTE. Care should always be individualized based on patient-specific and disease-specific considerations. The level of certainty in this table (low, moderate, or high) is based on author consensus regarding the quality of supporting evidence. See the text for details.
Abbreviations: AEs, adverse events; ASCT, autologous stem-cell transplantation; IRR, infusion-/injection-related reactions; mAb, monoclonal antibody; MM, multiple myeloma; NDMM, newly diagnosed multiple myeloma; OS, overall survival; PFS, progression-free survival; PFS2, progression-free survival from entry to second relapse; PO, oral; PR, partial response; QOL, quality of life; RCT, randomized controlled trial; RRMM, relapsed/refractory multiple myeloma; VGPR, very good partial response.

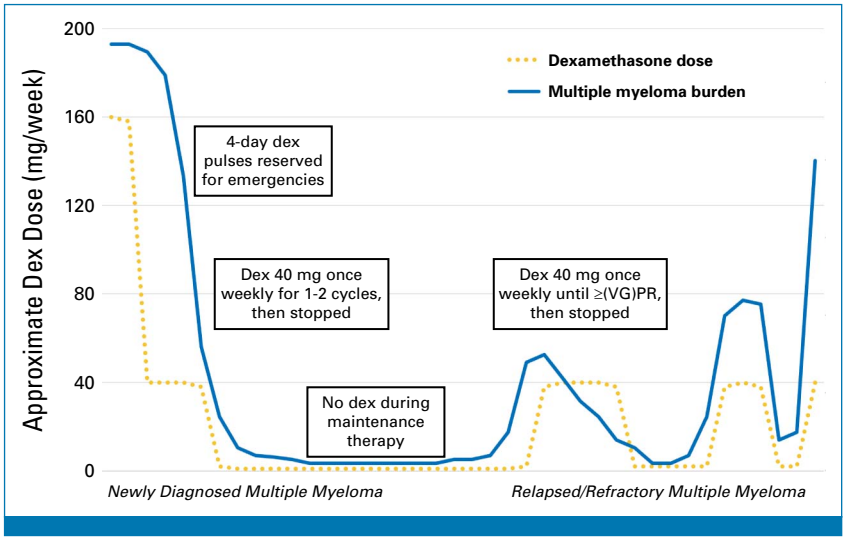


**FIG 4.** Suggested approach to dexamethasone dosing in multiple myeloma. Lower starting doses, for example, 20 mg once weekly, are reasonable to use in certain cases (eg, age ≥70 years, frailty, and/or comorbidities such as diabetes). Dexamethasone is not required beyond 1-2 cycles as a premedication for daratumumab or isatuximab (see the text for details). For medications such as carfilzomib or selinexor, dexamethasone 4-8 mg with each dose can often suffice as a premedication. dex, dexamethasone; mg, milligrams; (VG)PR; partial response/very good partial response.

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frank Cushingoid features if they remained alive.[19] Further research is needed to validate the efficacy of corticosteroid-sparing treatment algorithms, particularly in settings beyond ASCT-ineligible NDMM. Similarly, future studies are needed to better characterize the impact of dexamethasone's many low-grade toxicities on patient QOL (eg, through the dedicated SSQ-MM patient-reported outcome measure)[4] and to optimize the drug's benefit-risk ratio during each phase of therapy. As this type of research expands in quantity and quality, we expect to see evidence-based "down with dex" strategies become the norm in plasma cell disorders rather than the exception.


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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

**Past, Present, and Future of Dexamethasone in Multiple Myeloma and AL Amyloidosis**

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